

Probability Mass Functions

CSU, Chico Math 350

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outline

Random Variables

Estimating Discrete Prob. Distributions w/Simulation

Sampling from a Theoretical PMF

References

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Definition

Given a random experiment with an outcome sample space of $\mathcal{S}$. A function that assigns one and only one real number to each element in $\mathcal{S}$ is called a **random variable**.

Example 1

1. Consider an experiment that is the single roll of a die, where the number of spots on the face up side of the die when rolled is observed.

Outcome space is $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$

Let the random variable X indicate the number of spots on the side facing up.

The space of the random variable X is then $\{1, 2, 3, 4, 5, 6\}$.

Example 2

2. Say a person has 2 dogs and 2 cats in their household, so $\mathcal{S} = \{\text{"cat"}, \text{"dog"}\}$. Let Y be a random variable that denotes the type of animal. Y then maps each element in $\mathcal{S}$ to one and only one real number:

Let $Y = 0$ denote a dog, and $Y = 1$ denote a cat.

When the sample space only has two outcomes, it is considered a binary random variable. Often the choice of numbers used for a binary random variable are 0 and 1.

NOTE: These often correspond to FALSE and TRUE, respectively.

Distribution of a Random Variable

When a probability distribution has been specified on the sample space of an experiment, we can determine a probability distribution for the possible values of each random variable X .

This section is focused on probability distributions for **discrete** distributions. It is said that X has a discrete distribution if X can only take the values of a finite number k different numbers $x_1, x_2, \dots, x_k$ or at most, an infinite sequence of different values $x_1, x_2, \dots$.

Discrete vs. Continuous Reminder

Random variables that can take any value in an interval are called **continuous** and will be discussed in a later chapter.

Working with discrete random variables requires summation while continuous random variables require integration.

Reminder: Discrete variables are integers and usually represent a count of something while continuous variables take values in an interval of real numbers and often measure something.

Definition: PMF

A discrete random variable is a variable that takes integer values and is characterized by a *probability mass function (PMF)*. The PMF p of a random variable X is given by:

$$p(x) = P(X = x)$$

The above equation can be read as: the probability that the random variable X is equal to some value, x .

PMF Properties

Properties that the PMF satisfies:

1. $p(x) \geq 0 \quad \forall x$
2. $\sum_x p(x) = 1$

The term *probability distribution* is a more generic term that describes the probabilities for each different value a random variable can take on. This holds for both discrete and continuous random variables.

We will use the term probability distribution for all random variables, but the PMF is specific to discrete random variables, and PDF (Chapter 4) is specific to continuous random variables.

Example: Fixed Die

Consider a fixed six-sided die where the cube is shortened in the one-six direction. This has the effect that 1's and 6's have a probability of $1/4$ of being rolled, where the other faces each have a probability of $1/8$.

x	1	2	3	4	5	6
$p(x)$	$1/4$	$1/8$	$1/8$	$1/8$	$1/8$	$1/4$

Is this a valid PMF, and why? Recall the properties we went over a few slides back...

Exercise: Two Dice

Suppose you roll 2 six-sided fair dice. Let X be the sum of the two dice. Write out the PMF. Don't forget you can refer to Example 2.8 in the textbook to visualize the sample space.

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Example: Two More Dice

Suppose your roll 2 six-sided fair dice. Let X be the sum of the two dice. We can simulate this with the following:

```
die <- 1:6 #standard 6-sided values
d1  <- sample(die, 1000, replace = TRUE) #1000 rolls
d2  <- sample(die, 1000, replace = TRUE) #1000 rolls
sum.2d6 <- d1 + d2 #add the two vectors
```

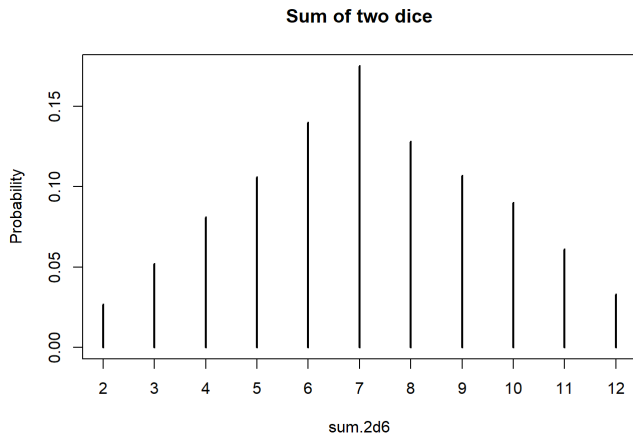
One possible estimate for the PMF of X is:

```
proportions(table(sum.2d6))
## sum.2d6
##      2      3      4      5      6      7
## 0.027 0.052 0.081 0.106 0.140 0.175
##      8      9     10     11     12
## 0.128 0.107 0.090 0.061 0.033
```

Plotting the PMF

We can use the function `plot` to plot the estimate of the PMF using the following code:

```
plot(proportions(table(sum.2d6)),  
     main="Sum of two dice", ylab="Probability")
```



Exercise 01: Coin Toss

Three coins are tossed and the number of heads X is counted. Write out the theoretical PMF for X first, then confirm via simulation.

x	0	1	2	3
$p(x)$				

Exercise 02: Urn the Third

Seven balls number 1-7 are in an urn. Two balls are drawn from the urn without replacement and the sum X of the number is computed. Estimate via simulation the PMF of X .

x	3	4	5	6	...
$p(x)$					

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Why Do This?

Sometimes you know what the theoretical pmf of a random variable is, but have need to draw samples from the known distribution. We can still use the `sample()` function to do so, we just provide it a vector of probabilities to use.

Example: Blood Types

In the United states, human blood comes in four types: O,A,B,AB.
Take a sample of thirty blood types with the following probabilities:
 $P(O) = 0.45$, $P(A) = 0.4$, $P(B) = 0.11$, $P(AB) = 0.04$

```
bloodtypes <- c("O","A","B","AB")
prob_bloodtypes <- c(0.45,.4,.11,.04)
sample_blood <- sample(bloodtypes, size = 30,
prob=prob_bloodtypes, replace=TRUE)
sample_blood[1:10] #quick peek to confirm

## [1] "A" "A" "O" "A" "O" "A" "B" "AB" "O" "O"
```

Estimated PMF:

```
proportions(table(sample_blood))
## sample_blood
##           A           AB           B           O
## 0.40000000 0.03333333 0.13333333 0.43333333
```

Exercise: Candy Colors

Suppose the proportion of M&Ms by color is: 14% yellow, 13% Red, 20% Orange, 12% Brown, 20% Green, and 21% Blue.

Answer the following questions using simulation:

- a) What is the probability that a randomly selected M&M is not green?
- b) What is the probability that a randomly selected M&M is red, orange, or yellow?

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References

references I

Speegle & Clair. Probability, Statistics, and Data: A Fresh Approach Using R, 2021.

Also adapted from notes by Dr. Roualdes, Dr. Donatello, and Dr. Gray.